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RESULTS

Results

01 · AVERAGE VARIANCE EXTRACTED (AVE)

convergent and discriminant validity

Table 1. Convergent validity

Construct	AVE	CR	ω	α
visual	.37	.61	.61	.63
textual	.72	.89	.89	.88
speed	.42	.69	.69	.69

Table 2. Discriminant validity (Fornell–Larcker)

	visual	textual	speed
visual	.61		
textual	.46	.85	
speed	.47	.28	.65

Fornell–Larcker: Diagonal = $\sqrt{\text{AVE}}$ (bold); off-diagonal = inter-construct latent correlations.

Table 3. Discriminant validity (HTMT)

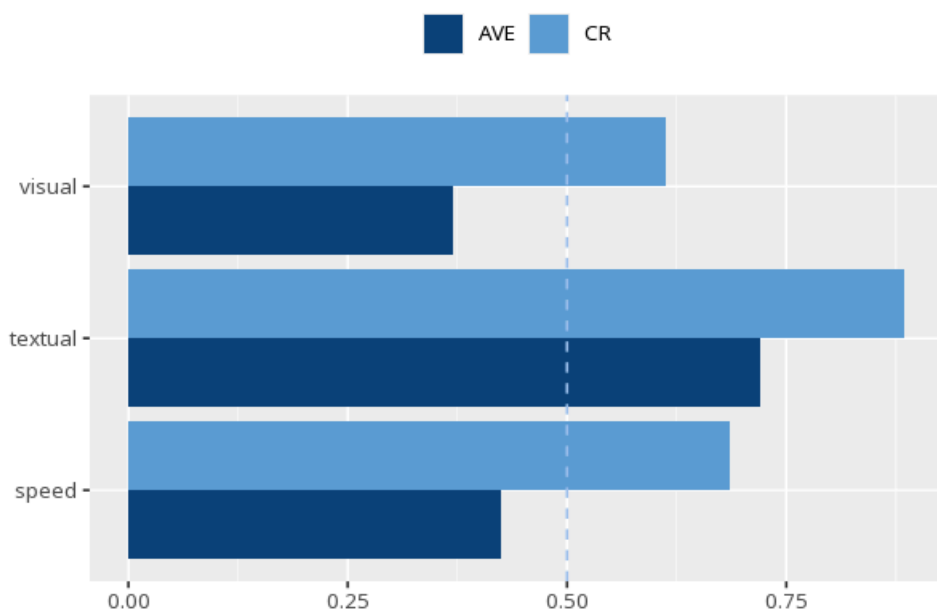
	visual	textual	speed
visual			
textual	.38		
speed	.39	.28	

Scope: When only 1 construct is defined, Tables 2 and 3 are suppressed (discriminant validity requires ≥ 2 constructs). Applies to reflective constructs only.

Cutoffs: AVE $\geq .50$ indicates adequate convergent validity, computed from the CFA loadings (Fornell & Larcker, 1981); CR $\geq .70$ is acceptable (Nunnally, 1978; Bagozzi & Yi, 1988); do not cite CR $\geq .70$ to Fornell & Larcker; HTMT $< .85$ indicates discriminant validity for conceptually distinct constructs (Henseler, Ringle & Sarstedt, 2015).

Reliability: ω (McDonald's) is the preferred reliability coefficient — model-based, not assuming tau-equivalence (McNeish, 2018); α is retained as a secondary/legacy column.

Figure. Validity overview



Understanding the numbers

AVE. The average share of a construct's item variance that it actually captures; $\geq .50$ (Fornell & Larcker, 1981) means the items converge well on their construct. Each row is one construct's own AVE; a construct below .50 may still be acceptable when its CR is above .60 (the Fornell & Larcker caveat), but treat that as a caution, not a lenient pass.

CR. Composite reliability - how reliably a construct's indicators measure it as a group; $\geq .70$ is acceptable (Nunnally, 1978; Bagozzi & Yi, 1988). Each row is one construct's own CR; a value below .70 flags a construct whose indicators may not hang together well.

ω (omega). The preferred headline reliability coefficient per construct - model-based, not assuming tau-equivalence (McNeish, 2018). Compare each construct's ω against its α in the same row - a large gap suggests the items' loadings are not equal.

α (alpha). Cronbach's alpha, shown per construct as a secondary/legacy coefficient alongside ω . Each construct's α is shown for reference alongside its own ω , the preferred coefficient.

How to read this test

AVE (Average Variance Extracted) is the average share of item variance a construct captures; $\geq .50$ means items converge on their construct (Fornell & Larcker, 1981). CR (Composite Reliability) $\geq .70$ indicates acceptable construct reliability. ω (McDonald's omega) is the preferred headline reliability coefficient; α (Cronbach's) is shown as a secondary/legacy coefficient. For discriminant validity, Fornell–Larcker (Table 2): each construct's $\sqrt{\text{AVE}}$ (bold diagonal) should exceed its correlations with other constructs. HTMT (Table 3): values $< .85$ support discriminant validity for conceptually distinct constructs — use HTMT as the primary criterion and Fornell–Larcker as a legacy check. AVE $< .50$ may still be acceptable when CR $> .60$ (F&L caveat), but treat this as a caution, not a lenient pass. All cutoffs are heuristics, not pass/fail gates. Applies to reflective constructs only.

APA template: Convergent validity was not fully supported, all AVE $\geq .50$ and CR $\geq .70$; discriminant validity held, all HTMT $< .85$.

Statistical basis: AVE (average variance extracted) convergent validity, $\geq .50$ (Fornell, C., Larcker, D. F. (1981). "Evaluating structural equation models with unobservable variables and measurement error." Journal of Marketing Research, 18(1), 39-50.); CR (composite reliability) $\geq .70$ acceptable (Nunnally, J. C. (1978). Psychometric Theory (2nd ed.). McGraw-Hill.); CR $\geq .70$ threshold, independent corroboration (Bagozzi, R. P., Yi, Y. (1988). "On the evaluation of structural equation models." Journal of the Academy of Marketing Science, 16(1), 74-94.); HTMT $< .85$ discriminant validity (primary criterion) (Henseler, J., Ringle, C. M., Sarstedt, M. (2015). "A new criterion for assessing discriminant validity in variance-based structural equation modeling." Journal of the Academy of Marketing Science, 43(1), 115-135.); Reliability preference (ω over α) (McNeish, D. (2018). "Thanks coefficient alpha, we'll take it from here." Psychological Methods, 23(3), 412-433.). Full references in the exported CITATIONS.txt.

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